Code The Change: Article/Info Retrieval
Title & Abstract Team

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Team Members

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Objective

Work with the Tackle Climate Change team to aid evidence synthesization using NLP methods

Problem

- ❏ Consolidating agroforestry data
- ❏ Papers are scattered and unstandardized
- ❏ Search process is tedious and inefficient

Solution

- ❏ Natural language processing model
- ❏ Input: title & abstract of a paper
- ❏ Output: 0 or 1 for relevance

Model Testing

Training Data

Manually labeled data from past reviews
~ 7,000 papers
~ 1,000 relevant

Text Vectorization

- Bag of words
- TF-IDF
- Word2vec
- BERT vectors
- SPECTER

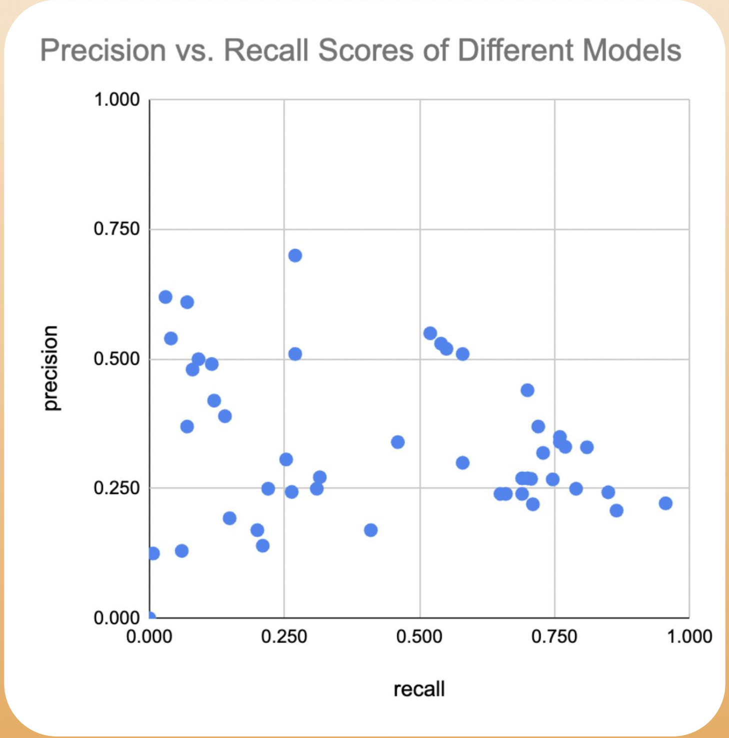
Model Type

- Linear Models
- Neural Networks
- Transformers

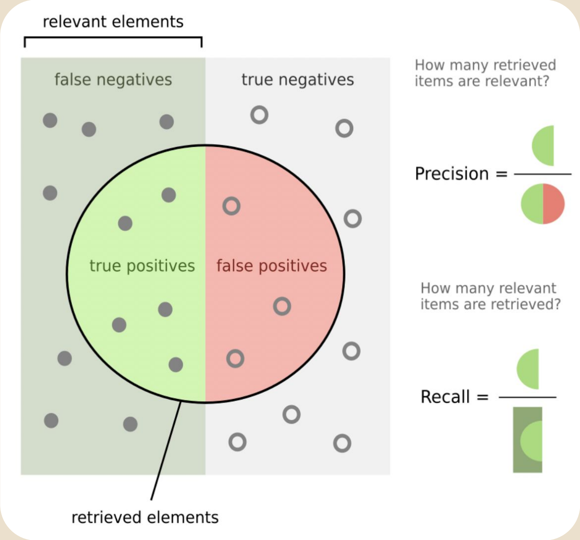
Hyperparameters

- Learning Rate
- Dropout Rate
- Batch Size
- etc.

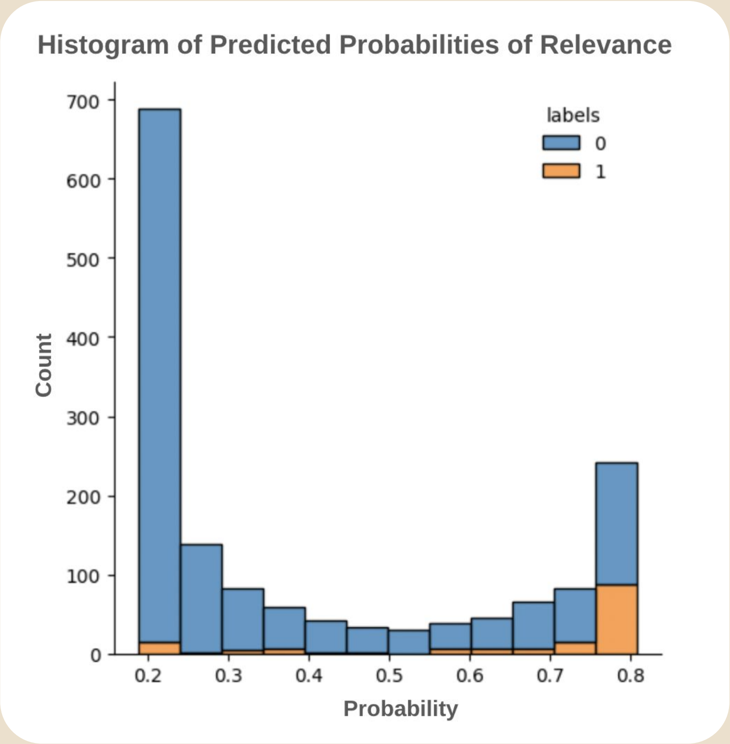
Model Performances



- Best recall: 0.96, best precision: 0.70
- Trade-off between precision and recall



A Closer Look: BERT-mini



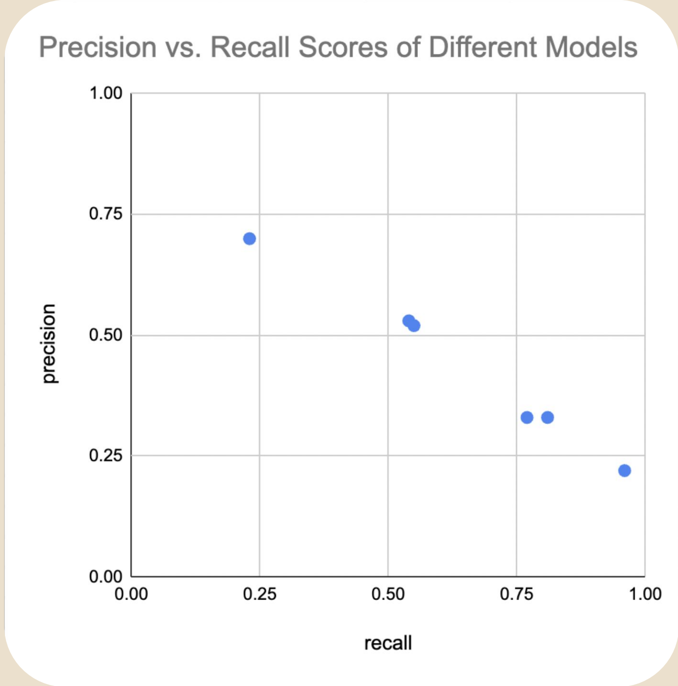
Notes

- Relevant → higher probability
- Irrelevant → lower probability
- A few relevant papers score very low and vice versa
- There are still more irrelevant papers than relevant papers in the largest bin

Key Takeaway

This model seems to recognize enough to follow the general pattern of relevance/irrelevance, but the quantity of irrelevant papers impair precision

Best Models: Pareto Optimal Front



		Precision	Recall
End-to-End Transformer Models	BERT precision	**0.70**	0.27
	BERT-mini	0.53	0.54
	SciBERT	0.52	0.55
Embedder + Neural Network	SPECTER NN 1 layer	0.33	0.77
	SPECTER NN 3 layers	0.22	**0.96**
Linear Model	Naive Bayes	0.33	0.81

Google Colab Demo

Code The Change

- Volunteer initiative based out of Harvey Mudd College
- Provide free technological solutions to non-profits
- Seeking new projects for the coming AY25-26 year

Next Steps

Please reach out to

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or check out

https://codethechange.cs.hmc.edu/

Questions?